

Copyright & Terms of Use

This resource package has been made available to you as part of the process of the Science by Doing website going offline.

After 14 January 2022 you may find that the information in the resources is missing, or the hyperlinks are broken or out-of-date. Science by Doing has no control over any outdated information or external links in these resources after this date.

The resources in this file are licensed under a **Creative Commons Attribution-NonCommercial-ShareAlike 4.0 International Public License (CC BY-NC-SA 4.0)**.

This means you are free to share and adapt all material, under the following terms:

Attribution: You must give appropriate credit and indicate if changes have been made.

NonCommercial: You may not use the material for commercial purposes.

ShareAlike: If you adapt the material, your contributions must also be distributed under a **CC BY-NC-SA 4.0** license.

For more information, visit the [Creative Commons license page](#) and the [Science by Doing Terms of Use](#).

Third Party Media Attributions

26 ga Copper Wire: Wormwould, Flickr, CC BY-NC 2.0, 3D Tin Can Phones: Chris Potter, StockMonkeys.com, Flickr, CC BY 2.0, Air: Thomas Hawk; Flickr; CC BY-NC 2.0, ANU group - BEC: Dr Tim Wetherell, ScienceWise Magazine, The Australian National University, Australian Mines Atlas: GeoScience Australia, CC BY 3.0 AU, Background water image: Spinks and Suns, Barium: Matthias Zepper, Wikimedia Commons, Blue dye in water: Domiriel, Flickr, CC BY-NC 2.0, Bose-Einstein Condensate (BEC): National Institute of Standards and Technology, Wikimedia Commons, Bose-Einstein Condensate: Professor Ken Baldwin, The Australian National University, Bromine: Dnn87, Wikimedia Commons, CC BY 3.0, Bronze age weapons Romania: Work of Romanian government. Wikimedia Commons, Brownian motion 5 Particles 150 frame: lookang, Francisco esquembre, Fu-Kwun, Wikimedia Commons, CC BY-SA 3.0, Brownian motion large: Francisco Esquembre, Fu-Kwun and lookang, Wikimedia Commons, CC BY-SA 3.0, Burning log: Anders Alexander, Andersofsydney, Flickr, CC BY 2.0, Buttons: Laineys Repertoire, Flickr, CC BY 2.0, Calcium in a protective argon atmosphere: Matthias Zepper, Wikimedia Commons, can_llauna_lata_#6: Remigi Buj, Flickr, CC BY-NC

2.0, Cesium: Dnn87, Wikimedia Commons, CC BY-SA 3.0, Chemistry Tutorial: Ice, Water & Steam: Atomic School, YouTube, <https://www.youtube.com/channel/UCiX8pAYWBpplbtUZTfGnRJw>, Chlorine2: Green horn 1, Wikimedia Commons, Chopping progression: Mike Hoff, Crashmaster 007, Flickr, CC BY-NC 2.0, Clouds: rocor, Flickr, CC BY-NC 2.0, CO2 emissions: Ian Britton, Flickr, CC BY-NC 2.0, Concrete: Nathan Gibbs; Flickr; CC BY-NC-SA 2.0, Constellation Crux: A. Fujii, Wikimedia Commons, CC BY 4.0, Cup of tea: soultea.de/André Helbig; Wikimedia, CC BY-SA 3.0, Do not paint: ASkaugset, Flickr, CC BY-NC 2.0, Does Australia expose children to unhealthy lead levels?: ABC News, Australian Broadcasting Commission (ABC), Dry ice held by tongs: Dennis Grice, Flickr, CC BY-NC 2.0, Dry ice in water: Dennis Grice, Flickr, CC BY-NC 2.0, EdibleSalt: Henningklevjer~commonswiki; Wikimedia; CC BY-SA 2.5, Effervescent tablet: Andrey Armyagov, Stock Photo, Image ID: 28566483, Einstein patent office: Lucien Chavan, Image cropped from original at the Historical Museum of Berne, Wikimedia Commons, Eisenerz mandelholz-harz hg: Hannes Grobe, Wikimedia Commons, CC BY 3.0, Elephant Toothpaste: Nicole Dupuis, YouTube, <https://www.youtube.com/watch?v=tnB-uU3w6g8&feature=youtu.be>, Example of Sublimation: Iodine: Wayne Breslyn, YouTube, <https://www.youtube.com/watch?v=Ft2KgtIP8Lk>, Extracting Metals and making Alloys: <http://www.bbc.co.uk/schools/gcsebitesize/science/aqa/metalsanduses/extractingmetalsrev7.shtml>, Farm plough: Alex Proimos, Alex E. Proimos, Flickr, CC BY-NC 2.0, Filtration: Matthew Myles, Flaming Flowers: Theen Moy; Flickr; CC BY-NC-SA 2.0, Fire conveyor belts: Antonis Lamnatos, Flickr, CC BY 2.0, Fraser Island S.S.Maheno Wreck: Majkelx, Wikimedia Commons, CC BY 1.0, Fuel for thought: Landline, Australian Broadcasting Commission (ABC), Gas Grill: Aldo Rado, Flickr, CC BY-NC 2.0, Goethite: Jake Slagle, Flickr, CC BY-NC 2.0, Gold bullion bars: Stevebidmead; Wikimedia; CC0 1.0, Grapes + Microwave = Plasma: Dr Derek Muller, Veritasium, Green goo: Jay Malone, Lunchbox Photography, Flickr, CC BY 2.0, Growing glaciers: Catalyst: Australian Broadcasting Commission (ABC), Helium balloons: Glen Van Etten, Flickr, CC BY-NC 2.0, Hematite: Stephanie Clifford, Sdixclifford, Flickr, CC BY 2.0, Henson Tires: cobalt123, Flickr, CC BY-NC 2.0, History of Australia's Minerals Industry: GeoScience Australia, CC BY 3.0 AU, How Stainless Steel Alloy is Formed?: International Stainless Steel Forum, How to make Graphene: Dr Derek Muller, Veritasium, Ice Cores: British Antarctic Survey, Natural History Museum, <https://www.youtube.com/watch?v=VjTsj-fi-p0&feature=youtu.be>, Ice on the moon: ABC Tasmania, Australian Broadcasting Commission (ABC), Ilod: Dnn87, Wikimedia Commons, CC BY 3.0, Indoor Air: Catalyst, Australian Broadcasting Commission (ABC), Iodine-evaporating: Jurii, Wikimedia Commons, CC BY 3.0, Jelly: Vladimir Agafonkin, Flickr, CC BY-NC 2.0, Klinzmoth, 1940's: Roadsidepictures, Flickr, CC BY-NC 2.0, Lead electrolytic and 1cm cube: Alchemist-hp; Wikimedia; CC BY-NC-ND

3.0, Lightbulb: Flickr, Lighting gas on a stove: Sunshinecity, Flickr, CC BY 2.0, Limonite US GOV: US Government, Wikimedia Commons, Liu Ding: Mountain, Wikimedia Commons, CC BY-SA 3.0, Magnetite Octahedron: Minas Gerals: Jake Slagle, Flickr, CC BY-NC 2.0, Making an Axe from Iron Ore: Trailer for the film Ore to Axe by Ken Koons: Ken Koons, YouTube, <http://youtu.be/vWxs7ZV5Ly8>, Man with shaving cream: Wikimedia Commons, Map of Australia: Google Maps, Melting butter: Andrew Malone, Flickr, CC BY 2.0, Melting caramel: Jacqueline, sweetbeetandgreenbean, Flickr, CC BY-NC 2.0, Mining boom set to turn to bust?: 7.30 Report, Australian Broadcasting Commission (ABC), Mining helps make your house: Copyright© 2020 OresomeResources.com, Mount Tom Price: Tom Durnin, Muesli: George Shuklin; Wikimedia; Public domain, Muesli texture: Luigi Chiesa; Wikimedia; CC BY 3.0, Nails: mekong_virus, Flickr, CC BY-NC 2.0, Nitrogen: Camilla Hoel; Flickr; CC BY-NC-SA 2.0, Nitrogen (7 N): Hi-Res Images of Chemical Elements; Wikimedia; CC BY 3.0, Northwest sky over Lake Pukaki.NZ: Bernard Spragg. NZ, Flickr, CC0 .10, Old Elkton Bridge: Brent Moore, SeeMidTN.com (aka Brent), Flickr, CC BY-NC 2.0, Ore, what is it good for?: ABC Splash, Australian Broadcasting Commission (ABC), Pakiri Beach: Natalia V, itravelnz, Flickr, CC BY 2.0, People in hazmat suits: US Navy, Wikimedia Commons, People pouring liquid nitrogen: Penn State, Flickr, CC BY-NC 2.0, Periodic Table: <http://www.ptable.com/#Writeup/Wikipedia>, Person wearing plastic Hazmat suit: Stuart Rankin: Flickr, CC BY-NC 2.0, Pineapple upside down cake: Catherine, Sailor Coruscant, Flickr, CC BY 2.0, Plasma: Ontario Science Centre, Plates: Matt Ephraim, Flickr, CC BY-NC 2.0, Potassium-2: Wikimedia Commons, CC BY 1.0, Pouring liquid mercury bionerd: Bionerd, Wikimedia Commons, CC BY 3.0, Quantum-atom optics: https://physics.anu.edu.au/ampl/research/quantum-atom-optics_grouping/, Rb5: Dennis "S.K", Wikimedia Commons, CC BY 3.0, Red blood cells: Stock Photo © Alexstar #1216582, Red ochre fish: Djulirri 60, Google Art Project, Wikimedia Commons, Reverse timelapse deepseafish icesculpture: Taco Koopmans, TTMedia, Rock, files, iron: Minerals Council of Australia, CC BY 3.0 AU, Rubber man: Spinks and Suns, Rusty car: Stephan Ridgway, sridgway, Flickr, CC BY 2.0, Safe Work Australia: <https://www.safeworkaustralia.gov.au/>, Sand footprint: Magnus Franklin, Flickr, CC BY-NC 2.0, Sandcastle with moat, seen on Morro Strand State Beach: Mike Baird, mikebaird, Flickr, CC BY 2.0, Scissors: James Bowe, Flickr, CC BY 2.0, Ship-propeller: US Department of Transportation, Wikimedia Commons, Snow plow 2: D Coetzee, Flickr, Source for the recommended daily intake of iron table: National Health and Medical Research Council, Springs 'N Stuff): Arbyreed, Flickr, CC BY-NC 2.0, States of Matter: Dr Derek Muller, Veritasium, STEEL: From Start to Finish: Alliance for American Manufacturing, YouTube, <https://www.youtube.com/watch?v=9l7JqonyoKA&feature=youtu.be>, Strontium: Matt hias Zepper, Wikimedia Commons, Stuff of Genius: The Evolution of

Rubber: <http://www.geniusstuff.com/videos/evolution-of-rubber-video.htm>, Sulfur sample: Ben Mills, Wikimedia Commons, Sweet opposition: LL Twistiti, Flickr, CC BY-NC 2.0, Tändsticka: Flickr, brandbild, CC BY 2.0, "Tates grass tree Xanthorrhoea semiplana tateana": Paul Asman and Jill Lenoble, Paul and Jill, Wikimedia Commons, CC BY 2.0, Tea_&_Milk-3: Kapil Dev Singh, Flickr, CC BY-NC 2.0, Top of Atmosphere: NASA Earth Observatory, Wikimedia Commons, Toxic poisoning: 7:30 Tasmania, Australian Broadcasting Commission (ABC), Unit Map image: Spinks and Suns, Washed and graded iron ore: Peter Craven, CDEGlobal, Flickr, CC BY 2.0, Water Cycle: Commonwealth Scientific and Industrial Research Organisation (CSIRO), Water: Andreas Levers, Flickr, CC BY-NC 2.0, Water drop 001: José Manuel Suárez; Wikimedia; CC BY 2.0, What is Dry Ice?: ImaginationStationOH, YouTube, <https://www.youtube.com/watch?v=oVmlAqwgIRo>, Why is ice slippery?: Dr Derek Muller, Veritasium.